Bessel's Equation :-

$$x^2 y'' + x y' + (x^2 - \nu^2) y = 0 \quad \text{about } x = 0 \, . \quad (1)$$

- check $x = 0$ is a regular singular point

$$P(x) = \frac{1}{x} \, , \quad Q(x) = \frac{x^2 - \nu^2}{x^2}$$

$$\lim_{x \to 0} x \cdot P(x) = \lim_{x \to 0} 1 = 1 \, , \quad \lim_{x \to 0} Q(x) \cdot \frac{1}{x^2} = \lim_{x \to 0} (x^2 - \nu^2) = -\nu^2 \, .$$

Let $y(x) = x^r \sum_{m=0}^{\infty} a_m x^m$ be a solution of (1)

$$y'(x) = \sum_{m=0}^{\infty} (m+r) a_m x^{m+r-1} \, , \quad y''(x) = \sum_{m=0}^{\infty} (m+r)(m+r-1) a_m x^{m+r-2}$$

- $x^2 y'' + x y' + (x^2 - \nu^2) y = 0$

∞ $m+r$ ∞ $m+r+2$

- $x^2y'' + xy' + (x^2 - \nu^2)y = 0$

$$\Rightarrow \sum_{m=0}^{\infty}(m+r)(m+r-1)a_m x^{m+r} + \sum_{m=0}^{\infty}(m+r)a_m x^{m+r} + \sum_{m=0}^{\infty} a_m x^{m+r+2}$$

$m+r+2 = r+k$

$\underline{m = k-2}$

$$-\nu^2 \sum_{m=0}^{\infty} a_m x^{m+r} = 0$$

- Coefficient of x^r (Indicial Equation)

$$r(r-1)a_0 + ra_0 - \nu^2 a_0 = 0 \Rightarrow a_0\left(r(r-1) + r - \nu^2\right) = 0$$

$$\because a_0 \neq 0 \Rightarrow r^2 - r + r - \nu^2 = 0$$

$$\Rightarrow r^2 = \nu^2 \Rightarrow \underline{\underline{r = \pm\nu}}$$

- Coefficient of x^{r+1}

$$r(r+1)a_1 + (r+1)a_1 - \nu^2 a_1 = 0 \Rightarrow a_1\left(r(r+1) + (r+1) - \nu^2\right) = 0$$

Either $a_1 = 0$ OR $r(r+1) + (r+1) - \nu^2 = 0$

$(r+1)^2 - \nu^2 = 0 \Rightarrow (r+1)^2 = \nu^2$

Either $a_1 = 0$ OR $r(r+1) + (r+1) - \nu^2$
$(r+1)^2 - \nu^2 = 0 \Rightarrow (r+1)^2 = \nu^2$
$\Rightarrow r+1 = \pm\nu$ Not true

we get $a_1 = 0$

- Coefficient of x^{r+k}, $k = 2, 3, 4, 5, \_\_\_$

$$(r+k)(r+k-1)a_k + (r+k)a_k + a_{k-2} - \nu^2 a_k = 0$$

$$a_k\left[(r+k)\left[r+k-1+1\right] - \nu^2\right] = -a_{k-2}$$

$$a_k = \frac{-a_{k-2}}{(r+k)^2 - \nu^2}$$

Case $r = \nu$

$$a_k = \frac{-a_{k-2}}{(\nu+k)^2 - \nu^2} = \frac{-a_{k-2}}{\cancel{\nu^2} + k^2 + 2k\nu - \cancel{\nu^2}}$$

$-a_{k-2}$ —

$$a_k = \frac{-a_{k-2}}{k(k+2\nu)}$$

$a_1 = 0$

$\underline{\underline{k=3}}$ $$a_3 = \frac{-a_1}{3(3+2\nu)} = 0 \quad , \quad \boxed{a_{2m+1} = 0 \quad \forall\, m}$$

$k = 2s$ $\longrightarrow$ coefficient of even powers of x.

$$a_{2s} = \frac{-a_{2s-2}}{2s(2s+2\nu)} = \frac{-a_{2s-2}}{2^2 \cdot s(s+\nu)}$$

$\underline{\underline{s=1}}$ $$a_2 = \frac{-a_0}{2^2 \cdot (\nu+1)}$$

$s=2$ $\quad -a_2 \quad = \quad a_0$

$\underline{\underline{s=2}}$ $$a_4 = \frac{-a_2}{2^2 \cdot 2\,(\nu+2)} = \frac{a_0}{2^4 \cdot 2\,(\nu+1)(\nu+2)}$$

$\underline{\underline{s=3}}$ $$a_6 = \frac{-a_4}{2^2 \cdot 3\,(\nu+3)} = \frac{-a_0}{2^6 \cdot 3!\,(\nu+1)(\nu+2)(\nu+3)}$$

$$a_{2s} = \frac{(-1)^s\, a_0}{2^{2s} \cdot s!\,(\nu+1)(\nu+2)\cdots(\nu+s)}$$

$\underline{\underline{\text{Case}}}$ $r = \nu = n$ (an integer, $\nu \geq 0$)

$$a_{2s} = \frac{(-1)^s\, a_0}{2^{2s}\, s!\,(n+1)(n+2)\cdots(n+s)},$$

$$\boxed{\text{we choose } a_0 = \frac{1}{2^n\, n!}}$$

$$a_{2s} = \frac{(-1)^s}{2^{2s+n}\, s!\; n!\,(n+1)(n+2)\cdots(n+s)}$$

$2s$

$$2^{2s+n}\, s!\; n!\,(n+1)(n+2)\ldots(n+s)$$

$$a_{2s} = \frac{(-1)^s}{2^{2s+n}\, s!\,(n+s)!}$$

$$y = x^r \sum_{m=0}^{\infty} a_m x^m$$

Solution of Bessel's Equation

$$y(x) = x^n \sum_{s=0}^{\infty} \frac{(-1)^s}{2^{2s+n}\, s!\,(n+s)!} x^{2s}$$

The above function is called Bessel's function of first kind of order n.

Case $\nu > 0$; ν may not be an integer

Gamma function :- $\Gamma(\nu) = \int_0^\infty e^{-t}\, t^{\nu-1}\, dt \quad (\nu > 0)$

∞

$$\Gamma(\nu+1) = \int_0^\infty e^{-t}\, t^{\nu}\, dt$$

$$= \underbrace{t^{\nu}\cdot(-e^{-t})\Big|_0^\infty}_{=0} - \int_0^\infty \nu\, t^{\nu-1}\cdot -e^{-t}\, dt$$

$$= \nu \int_0^\infty e^{-t}\, t^{\nu-1}\, dt = \nu\Gamma(\nu)$$

$$\boxed{\Gamma(\nu+1) = \nu\Gamma(\nu)}$$

$$\Gamma(1) = \int_0^\infty e^{-t}\, dt = -e^{-t}\Big|_0^\infty = 1$$

$$\Gamma(n+1) = n!$$

$$\Gamma(2) = 1\Gamma(1) = 1$$

$$\Gamma(3) = 2\Gamma(2) = 2 \qquad \Gamma(4) = 3\Gamma(2) = 3\times 2 = 3! \qquad \Gamma(5) = 4\Gamma(3) = 4!$$

$(\quad)^8 \quad a_n$

$$a_{2s} = \frac{(-1)^s \, a_0}{2^{2s} \, s! \, (\nu+1)(\nu+2) \cdots (\nu+s)}$$

If ν is not an integer, we choose $a_0 = \dfrac{1}{2^{\nu}\,\Gamma(\nu+1)}$

$$a_{2s} = \frac{(-1)^s}{2^{2s+\nu} \, s! \, \Gamma(\nu+1)\left[(\nu+1)(\nu+2) \cdots (\nu+s)\right]}$$

$\Gamma(\nu+2)$

$\Gamma(\nu+3)$

$$a_{2s} = \frac{(-1)^s}{2^{2s+\nu} \, s! \, \Gamma(\nu+s+1)}$$

Solution of Bessel's Equation

$$J_\nu(x) = x^{\nu} \sum_{s=0}^{\infty} \frac{(-1)^s \, x^{2s}}{2^{2s+\nu} \, s! \, \Gamma(\nu+s+1)}$$

- Bessel's function of first kind.

* If ν is not an integer

$$J_{-\nu}(x) = x^{-\nu} \sum_{s=0}^{\infty} \frac{(-1)^s \, x^{2s}}{2^{2s-\nu} \, s! \, \Gamma(s-\nu+1)}$$

* $J_\nu(x)$ and $J_{-\nu}(x)$ are linearly independent function. Then complete solution of Bessel's Equation is

$$y(x) = C_1 J_\nu(x) + C_2 J_{-\nu}(x).$$

$J_{-n}(x) = (-1)^n J_n(x)$

- If $\nu = n$ (an integer), $J_{-n}(x) = (-1)^n J_n(x)$,

i.e. $J_n(x)$ and $J_{-n}(x)$ are <u>L.D.</u>

$$J_{-n}(x) = x^{-n} \sum_{s=0}^{\infty} \frac{(-1)^s \, x^{2s}}{2^{2s-n} \, s! \, \Gamma(s-n+1)}$$

$\Gamma(n) = \infty$ if $n \leq 0$

$\Gamma(s-n+1) = \infty$ if $s-n+1 \leq 0$

$\Rightarrow s \leq n-1$

$\Rightarrow$ in $J_{-n}(x)$, coefficient x^{2s-n} for $0 \leq s \leq n-1$ is <u>zero.</u>

$$J_{-n}(x) = x^{-n} \sum_{s=n}^{\infty} \frac{(-1)^s \, x^{2s}}{2^{2s-n} \, s! \, \Gamma(s-n+1)}$$

$$J_{-n}(x) = x \sum_{s=n} \frac{}{2^{2s-n}\, s!\, \Gamma(s-n+1)}$$

Replace $s+n=s$

$$J_{-n}(x) = x^{-n} \sum_{s=0}^{\infty} \frac{(-1)^{s+n}\, x^{2(s+n)}}{2^{2(s+n)-n}\,(s+n)!\,\Gamma(s+\cancel{n}-\cancel{n}+1)}$$

$$J_{-n}(x) = x^{n}(-1)^{n} \sum_{s=0}^{\infty} \frac{(-1)^{s}\, x^{2s}}{2^{2s+n}\,(s+n)!\, s!} = (-1)^{n} J_{n}(x)$$

<u>Case I</u> <u>If $v=n$ is an integer</u>

(a) $v=0$ (Repeated roots)

$$y_2(x) = y_1(x)\ln x + \sum_{k=1}^{\infty} b_k x^{k}$$

$$y_2(x) = y_1(x) \ln x + \sum_{k=1} \ldots {}_k$$

$$y_2(x) = J_0(x) \ln x + \sum_{k=1}^{\infty} b_k x^k$$

$$y_2(x) = J_0(x) \ln(x) + \sum_{k=1}^{\infty} \frac{(-1)^{k-1}\left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k}\right)}{2^{2k} \cdot k!} x^{2k}$$

(b) $\nu = n$ (an integer but $\nu \neq 0$)

$$Y_n(x) = \frac{1}{\sin(n\pi)} \left(J_n(x) \cos(n\pi) - J_{-n}(x) \right)$$

The above function is called Bessel's function of second kind of order n.

of second kind of order n.